

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE-6003-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Microprocessor and Microcontrollers

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Draw the 8086 pin diagram. Explain the functionality of each pin.
b) What are the interrupts available in 8086? Give the interrupt vector table of 8086 microprocessor.
2. a) Write a program to add two 8-bit BCD number in the 8086 and store the result in memory.
b) Explain the different modes of operation of PPI 8255.
3. a) Explain architecture of programmable peripheral interface chip 8255.
b) Discuss how 8254 is used?
 - i) To generate delay
 - ii) To generate square wave
4. a) Describe the importance of DMA scheme. Draw the block diagram and explain the operation of 8257DMA controller.
b) Explain the block diagram and function of each block of 8251 USART.

EE-6003-CBGS

PTO

[2]

5.
 - a) Draw the pin diagram of 8051 microcontroller and explain the function of each pin.
 - b) Explain the matrix keyboard interface of 8051.
6.
 - a) Explain data types and subroutines. List various application of 8051.
 - b) Draw the block diagram and explain architecture of 8051 microcontroller.
7.
 - a) Explain the interfacing of stepper motor with 8051 microcontroller.
 - b) Design 8051 microcontroller based firing circuit.
8. Explain the following.
 - a) Instruction set of 8086.
 - b) Functional block diagram of 8096 microcontroller.
